

Taxonomy of Multimodal Time Series Models

Multimodal Time Series Paradigm (Data, Fusion, Role, Task, Domain)

1. Modality Pairing

- TS + Text (Reports, News, Prompts)
- TS + Vision (Plots, Spectrograms, MAE)
- TS + Audio (Speech AMs, Bioacoustics)
- TS + Vision + Text (Tri-modal VLM, EHR)
- TS + Multi-Rate Signals (100Hz/Hourly)
- TS + Continuous Wearables (SOTER)

2. Fusion Architecture

- Patch Reprogramming (Time-LLM, OFA)
- Visual Mamba / 2D Vim (TriTS, VisionTS)
 - Delay-Aware Dual-Path SSM (DeMa)
- Continuous Neural CDE / ODE (SOTER)
- Cross-Attention Adapter (Time-VLM, MedFuse)
 - Semantic-Spline SSM (ss-Mamba)

3. Non-TS Role

- Auxiliary Context / Condition
- Conformal Calibration Anchor (Achour)
- Conversational & Reasoning Interface
 - Metric Alignment Target (TRACE)
- Dynamic Red-Teaming Contamination Defense
 - Autonomous Agent Tool Dispatching

4. Downstream Tasks

- Multimodal Forecasting (Point / Conformal)
 - Anomaly Detection & Imputation
 - Time Series QA & Reasoning
- Cross-Modal Retrieval (TRACE, Any2Any)
 - Wearable Bio-Signals (SOTER)
- Earth System Weather / Downscaling

Application Domains: Healthcare & EHR • Meteorology & Climate • Finance & Markets • Energy & Smart Grids • Geophysics & Bioacoustics